

Batch 1 Foundation Implementation

Implemented the first v0.7 curriculum batch in the actual app: the foundation story from what an LLM is through training, pretraining, overfitting, fine-tuning, and alignment.

7

Batch 1 lessons

2

new Journey cards

28

total lessons

0

new games

What Changed

- Rewrote Batch 1 lesson data with lifecycle fields, durable/temporary distinctions, prompt/response notes, misconceptions, and feedback.
- Renamed the existing `history` lesson to Rationalists vs Empiricists while preserving progress compatibility.
- Added Overfitting and Generalization plus Alignment as new Journey lessons.
- Added lightweight React/SVG visual aids for Batch 1 concepts.
- Expanded glossary vocabulary and related-term links for training, overfitting, fine-tuning, and alignment.
- Updated review route, lesson-card export target, and visible app version.

Files Changed

- `package.json`
- `src/components/VisualAids.tsx`
- `src/data/content.ts`
- `src/data/contentReview.js`
- `src/data/exercises.ts`
- `src/main.tsx`
- `docs/REVIEW_NOTES.md`
- `docs/curriculum/BATCH_1_IMPLEMENTATION_REPORT_V0_8.md`
- `docs/curriculum/BATCH_1_REVIEW_MATRIX_V0_8.md`
- `docs/curriculum/SOURCE_NEEDS_V0_8_BATCH_1.md`

- docs/content-inventory/prompt-life-lesson-cards-v0-8-batch-1.pdf
- docs/screenshots/v0-8-batch-1-*.png

Verification

- npm install : passed.
- npm run typecheck : passed.
- npm run build : passed.
- npm run export:lesson-cards : passed.
- 390px and 320px overflow checks passed for the first seven lessons.
- Reset QA passed in an isolated browser session.

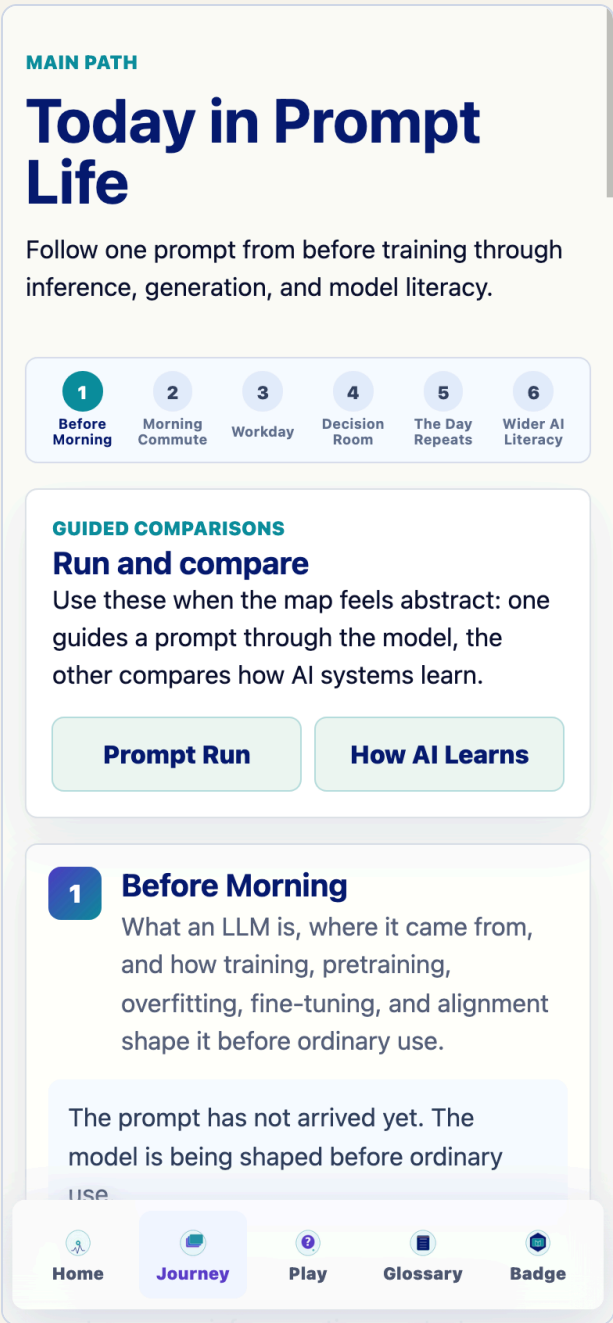
Known Issues

- The `history` id is intentionally preserved even though the visible title changed.
- Batch 1 source review is still needed before publication-final copy.
- Batch 1 uses existing micro-interactions; no new exercise types were added.

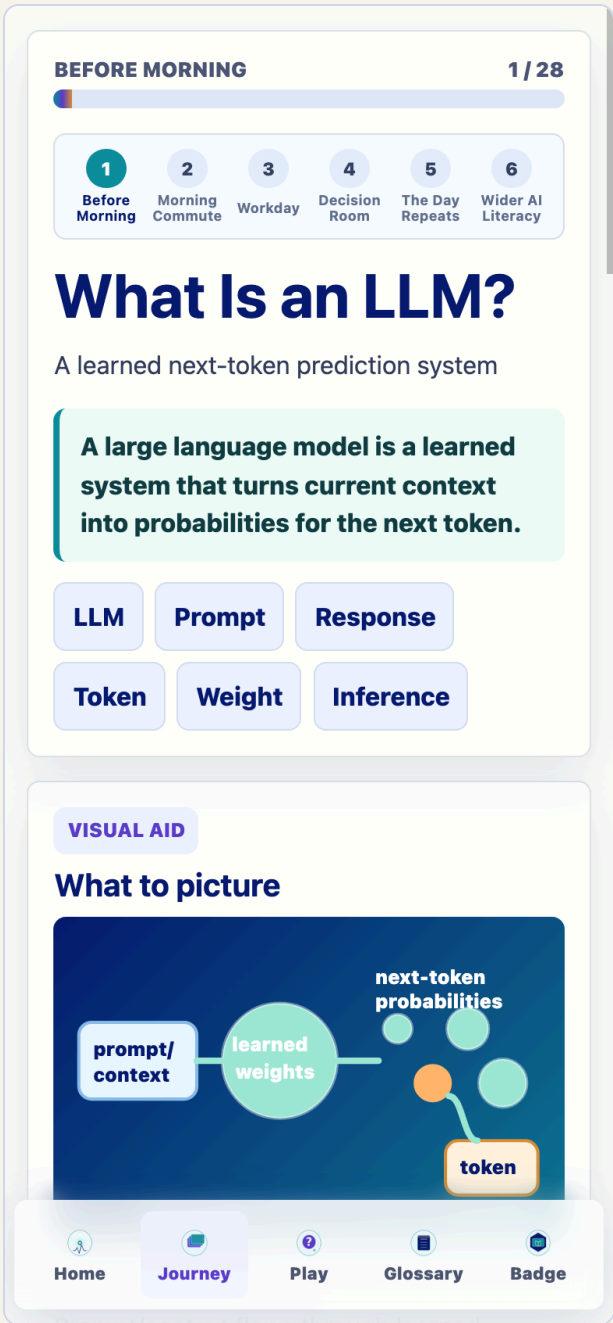
Next Steps

1. Implement Batch 2: Inference, Prompt vs Response, Tokenization, Token IDs, Embeddings, Vectors, and Tensors.
2. Add a source/bibliography pass for Batch 1 claims.
3. Add automated route checks for the first seven lessons, review route, glossary drawer, reset, and Badge denominator.

Screenshots



Journey top with Batch 1 foundation section.



What Is an LLM lesson top.

BEFORE MORNING

2 / 28

1

Before Morning

2

Morning Commute

3

Workday

4

Decision Room

5

The Day Repeats

6

Wider AI Literacy

Rationalists vs Empiricists

Rules versus learned patterns

Symbolic AI tries to reason with explicit rules and symbols; deep learning learns useful patterns from data by adjusting weights.

Rationalism

Empiricism

Symbolic AI

Deep learning

Training

Weight

VISUAL AID

What to picture

symbolic rationalist

deep learning empiricist

Home

Journey

Play

Glossary

Badge

Rationalists vs Empiricists replaces the old Two AI Traditions title.

VISUAL AID

What to picture

Training Loop

Predict, compare, loss, update weights, repeat. Training changes weights.

CORE IDEA

Core idea

During training, the model sees examples, predicts something, measures how wrong it was, and updates its weights to do slightly better next time. This happens over many examples and many updates. Training is where durable model capability is created.

MODEL LIFECYCLE

training before ordinary use

WHERE IT HAPPENS

Home

Journey

Play

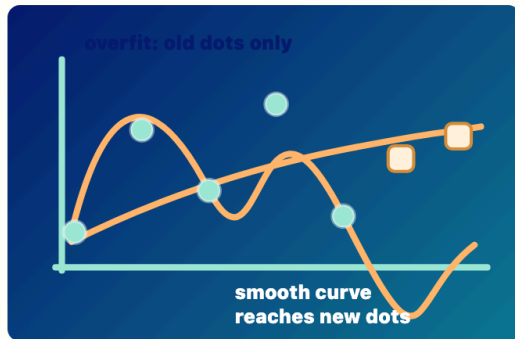
Glossary

Badge

Training visual: predict, compare, loss, update weights, repeat.

VISUAL AID

What to picture



Overfitting vs Generalization

Memorizing examples is not the same as learning transferable patterns.

CORE IDEA

Core idea

A model can fit old examples very well but still fail on new cases. That is overfitting. Good learning captures patterns that transfer. For LLMs, generalization is why a model can answer prompts it has never seen exactly before. Overfitting is why evaluation needs unseen examples, not just training performance.

MODEL LIFECYCLE

training before ordinary use

Home Journey Play Glossary Badge

Overfitting visual: old dots versus transferable pattern.

STILL BEFORE MORNING

7 / 28

1

Before Morning

2

Morning Commute

3

Workday

4

Decision Room

5

The Day Repeats

6

Wider AI Literacy

Alignment

Shaping behavior toward human goals

Alignment is the effort to shape AI behavior toward human instructions, preferences, safety boundaries, and values.

Alignment

Instruction tuning

Human feedback learning

RLHF

Preference optimization

Policy

Guardrail

Evaluation

Fine-tuning

VISUAL AID

What to picture

Home Journey Play Glossary Badge

Alignment lesson top with glossary chips wrapping cleanly.

BRAIN BRIDGE

Helpful, then limited

HELPFUL COMPARISON

Like learning social norms or professional expectations.

WHERE IT BREAKS

The model does not acquire human values, moral agency, conscience, or care. Its behavior is shaped by training and system constraints.

TINY INTERACTION

Trace the guardrails

FINE-TUNING PATH

Nudge path



broad model

Fine-tuning adds targeted training that nudges behavior without starting over.



Home



Journey



Play



Glossary



Badge

CHECKPOINT

What does alignment try to do?

A

Shape model behavior toward human goals, instructions, and safety boundaries

B

Make the model conscious

C

Guarantee every answer is true

D

Permanently solve all AI risk

Insight unlocked. Alignment shapes behavior, but it is not magic morality.

REFLECT

Teach it back in one sentence.

Example: Inference uses the model's weights but does not rewrite them.

Next lesson



Home



Journey



Play



Glossary



Badge

Brain Bridge explicitly limits the cognition metaphor.

Checkpoint correct feedback.

CHECKPOINT

What does alignment try to do?

- A** Shape model behavior toward human goals, instructions, and safety boundaries
- B** Make the model conscious
- C** Guarantee every answer is true
- D** Permanently solve all AI risk

Not quite. Alignment is a set of technical and human processes, not consciousness or a perfect safety guarantee. Choose another answer to retry.

REFLECT

Teach it back in one sentence.

Example: Inference uses the model's weights but does not rewrite them.

Home

Journey

Play

Glossary

Badge

PROMPT LIFE V0.8.0

Lesson Cards

One accessible curriculum-review card per lesson, built from current app data plus inventory notes and rubric scores.

1 / 28

BEFORE MORNING

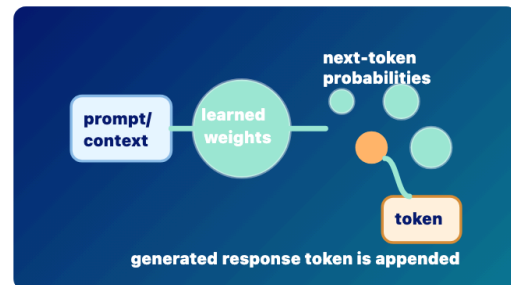
What Is an LLM?

A learned next-token prediction system

Stage: architecture

Priority: low

Rubric avg: 4.4/5



Prompt to Prediction

Prompt/context flows through learned weights into next-token probabilities; one generated response token is appended.

DEFINITION

A large language model is a learned system that turns current context into probabilities for the next token.

Checkpoint wrong feedback.

Review route and PDF source surface.